Stochastic Simulation Analysis: Dynamics of an EV Charging Station

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1 Introduction

This report describes a discrete-event simulation (DES) of an electric vehicle (EV) charging station with four identical chargers. The model includes time-varying arrivals and service times that depend on each vehicle's battery level. To better reflect real-world behavior, vehicles may leave if they wait too long or if the queue becomes too crowded. The system is evaluated based on 800 completed charging sessions.

2 System Model and Formalization

2.1 State Representation

The system state at any time $t \geq 0$ is defined by the state vector $\mathcal{S}(t) = \{Q(t), S(t)\}$, where $Q(t) \in \mathbb{N}_0$ represents the number of vehicles awaiting service in a first-come, first-served (FCFS) queue, and $S(t) \in \{0, \ldots, 4\}$ denotes the number of busy charging points. In addition to these primary variables, the simulation tracks cumulative counters for total arrivals (A_{count}), reneged vehicles (R_{count}), early departures (E_{count}), and total completions (C_{count}).

2.2 Performance Metrics

To assess the efficiency and service quality of the station, we define five key performance indicators calculated over the interval $[0, T]$, where T is the time of the N-th completion. The average queue length is computed as a time-weighted integral $\bar{L}_Q = \frac{1}{T}\int_0^T Q(t)dt$, while the charger utilization is normalized across all charging stations as $U = \frac{1}{4T}\int_0^T S(t)dt$. For vehicles that successfully transition to a Charging Station, the average waiting time is given by $\bar{W} = \frac{1}{M}\sum_{i=1}^{M}(t_{start,i} - t_{arrival,i})$. System loss and behavioral intensity are measured by the reneging fraction $F_R = R_{count}/A_{count}$ and the early-departure fraction $F_E = E_{count}/C_{count}$, respectively.

3 Operational Dynamics and Event Logic

The simulation is based on four primary events that manipulate the Future Event List (FEL) and update the system state.

Arrival Process Upon the n-th arrival at time t, the system calculates a deterministic battery level $b_n = 0.5 \cdot |\sin(n\pi/7) + 1|$ and schedules the next arrival at $t + \Delta t_n$, where $\Delta t_n = 15(1 + \sin(n\pi/12))^2 + 2$. If $S(t) < 4$, the vehicle immediately occupies a charger, setting $S(t) \leftarrow S(t)+1$ and scheduling a Departure Event at $t + 60(1 - b_n)$. Conversely, if all chargers are occupied, the vehicle enters the queue. At this moment, a Reneging Event is scheduled at $t + \tau_n$, where

$\tau_n = 20(1+|\cos(n \cdot e)|)$ represents individual patience. Furthermore, if $Q(t)$ reaches a multiple of 5, the "Queue Pressure" logic is invoked: each currently charging vehicle with a remaining time exceeding 15 minutes may independently schedule an Early Departure for $t+2$ with probability $p = 0.2$.

Departure and Early Departure Mechanics A Departure Event signifies the opening of a charging station. The system increments C_{count} (and E_{count} if the departure was premature) and decrements $S(t)$. If $Q(t) > 0$, the FCFS discipline is applied: the head-of-line vehicle is removed from the queue, its associated Reneging Event is purged from the FEL, and a new Departure Event is scheduled. This sequence ensures that the transition from queuing to service is consistent.

Reneging Dynamics A Reneging Event occurs if a vehicle's patience τ_n expires before a Charging Station becomes available. The logic involves decrementing $Q(t)$ and incrementing R_{count}. Crucially, if the vehicle reaches the Charging Station before this time, the event must be explicitly cancelled to prevent erroneous state updates.

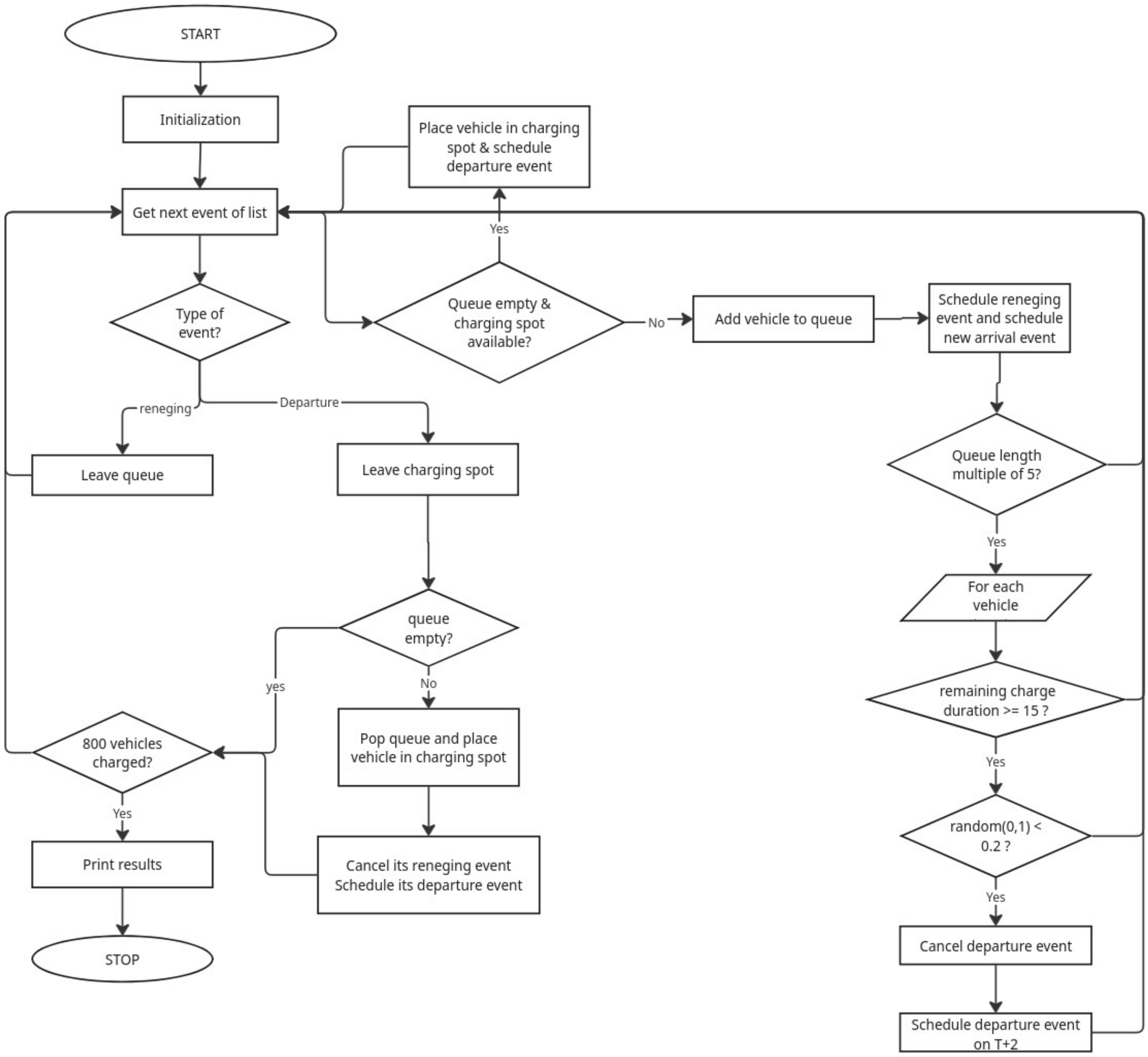

Figure 1: Flowchart of the Simulation Model

4 Event List Interaction and State Tracking

The accuracy of the simulation depends on careful management of the Future Event List (FEL). Reneging and service events are coordinated through event cancellation: when a vehicle starts charging, its scheduled reneging event is removed. The early departure mechanism works as an override, if a vehicle is selected to leave early, its original completion event is replaced with a high-priority event scheduled at time $t + 2$. At each event time, the accumulated area under $Q(t)$ and $S(t)$ is updated before any changes are made to the state variables.

5 Results and Discussion

Table 1: Performance Metrics for $N = 800$, Seed 70

Metric	**Value**
Termination Time (T)	21152.30 min
Average Queue Length ($\bar{L}_Q$)	0.26 vehicles
Average Waiting Time ($\bar{W}$)	15.90 min
Reneging Fraction (F_R)	0.06
Charger Utilization (U)	0.28
Early-Departure Fraction (F_E)	0.04

The results show that the early departure mechanism helps stabilize the system. When the queue becomes long ($Q \in \{5, 10, \dots\}$), some vehicles leave early, which increases turnover. This reduces the average queue length and keeps the reneging fraction at about 6%. However, this comes with a trade-off: improving flow and reducing waiting times means some charging sessions are not completed. Without this mechanism, peak arrival periods would likely cause more congestion and a higher number of vehicles leaving due to long waits.